

Intrinsic Motivation for Exploration in Sparse-Reward MiniGrid

Results and slide-organisation notes (response to docs/SPEC.md)

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation — RND vs. LPM under PPO

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Abstract

After the MiniWorld 3D-maze reproduction showed that *coverage* fails to separate exploration methods in an extrinsic-free maze, we move to **sparse-reward MiniGrid**, where a real task reward lets intrinsic motivation actually pay off. Under a common PPO agent (MLP on the 7×7 symbolic view) we compare a plain baseline (NONE), entropy-bonus exploration (ENTROPY), Random Network Distillation (RND) and the paper’s Learning Progress Monitoring (LPM), across a three-rung difficulty ladder (easy DoorKey-5x5, medium FourRooms, hard MultiRoom-N6), clean and under observation noise, with 3 seeds. Three headline results. **(1)** The intrinsic mixing coefficient β matters enormously and is environment-dependent: $\beta=0.05$ *drowns* the sparse reward (return $\rightarrow 0$); the usable range is ~ 0.001 – 0.005 . **(2)** Intrinsic motivation’s benefit is *difficulty-gated*: unnecessary on easy/medium (the PPO baseline already solves/explores them), but **decisive on hard MultiRoom-N6, where RND solves the task (eval return 0.32) while the baseline and entropy never reach the goal (0.0)**. RND beats LPM on clean hard exploration: RND’s always-positive novelty bonus drives directed exploration, whereas LPM’s signed learning-progress signal nets ≈ 0 per episode and under-explores. **(3)** That trade-off reverses under noise: on noisy DoorKey-5x5 **RND collapses** ($0.82 \rightarrow 0.035$) **while LPM stays robust** ($\rightarrow 0.95$) — a clean MiniGrid reproduction of the LPM paper’s noise-robustness claim. We give the figures, tables and agent traces, and map each to the planned slide deck.

1 Setup

Agent. PPO (Stable-Baselines3), MLP policy, 8 vectorised envs. Observation: `ImgObs` — the agent’s $7 \times 7 \times 3$ egocentric symbolic view, flattened to **147** dimensions. *We deliberately dropped the default `FlatObs` (2835-dim): $\sim 95\%$ of it is a constant one-hot of the (fixed) mission string, i.e. uninformative padding; `ImgObs` keeps the identical spatial information $\sim 19\times$ lighter.*

Exploration methods. (i) NONE: plain PPO, `ent_coef=0`. (ii) ENTROPY: PPO with `ent_coef=0.01` — the *non-intrinsic* (policy-stochasticity) comparator. (iii) RND: bonus = $\|f_\theta(s) - \hat{f}(s)\|^2$ (predictor vs. fixed random target). (iv) LPM: paper-faithful log-space reward $r^i = g_\phi - \log \text{MSE}$, gated until the error buffer fills. Intrinsic reward enters as $r = r^{\text{ext}} + \beta r_{\text{norm}}^i$.

Environments (difficulty ladder, one per tier). easy DoorKey-5x5 (subgoal shaping disabled $\rightarrow$ purely sparse), medium FourRooms, hard MultiRoom-N6. *KeyCorridor-S3R3 was tried as the hard rung and abandoned: it requires memory (carry a key, remember it, open a door), which a feed-forward MLP on the image-only view cannot represent — all methods scored 0 even at 3M steps. MultiRoom-N6 is memory-free (doors just toggle) so it is MLP-solvable, but exploration-hard.*

Metrics, seeds, budget. Final deterministic eval return (the sparse task reward; the eval env carries no intrinsic bonus), the per-episode extrinsic/intrinsic split, and sample-efficiency curves. 3 seeds, mean $\pm$ std. Budgets: easy/medium 1M, hard 2M PPO steps.

2 β : how to mix extrinsic and intrinsic reward (RQ2)

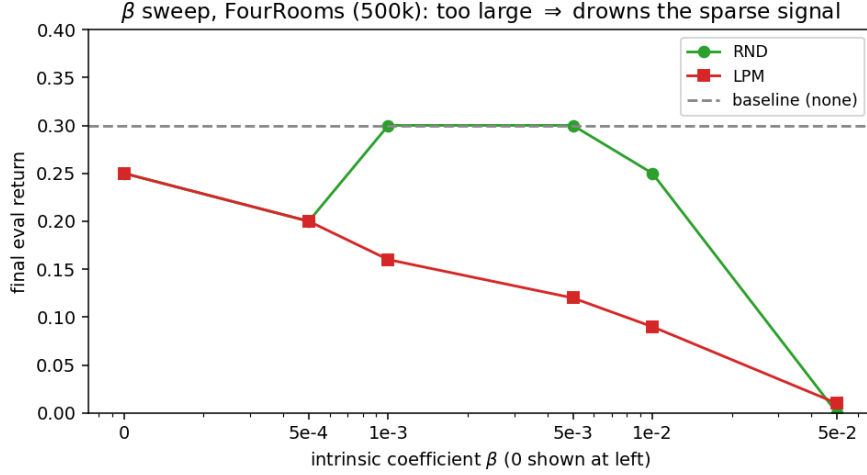


Figure 1: β sweep on FourRooms (500k). RND rises to a sweet spot ($\beta \approx 0.001$ – 0.005 , matching the baseline) then collapses to 0 at $\beta=0.05$; LPM declines monotonically.

At $\beta=0.05$ the per-episode *intrinsic* reward was 100–700 $\times$ the sparse extrinsic reward, so the policy optimises novelty and never learns the goal (eval $\rightarrow 0$). The usable band is ~ 0.001 – 0.005 . On the *hard* env a slightly larger β is needed to drive exploration, but too large still prevents convergence to exploitation: RND on MultiRoom-N6 only converges at $\beta=0.005$ (eval 0.28); at $\beta=0.01/0.05$ it reaches the goal during training but never settles into a reliable deterministic policy (eval ≈ 0). **Takeaway:** β is environment-dependent — small when intrinsic reward is a nuisance, larger when it is essential, with a sweet spot in between. *Slide 3.3.*

3 Does intrinsic reward beat the baseline? (RQ3)

env (clean)	NONE	ENTROPY	RND	LPM
DoorKey-5x5 (easy)	0.93	0.96	0.82	0.61
FourRooms (medium)	0.34	0.30	0.30	0.25
MultiRoom-N6 (hard)	0.00	0.00	0.32	0.00

Table 1: Final eval return (mean, 3 seeds). **Intrinsic motivation is difficulty-gated.**

Intrinsic motivation’s benefit appears *only* when exploration is genuinely hard. On easy and medium tasks the PPO baseline explores adequately on its own and any bonus slightly degrades it; on hard MultiRoom-N6 the baseline and entropy never reach the goal, while **RND solves it reliably** (0.32, all seeds). *Slide 3.4.*

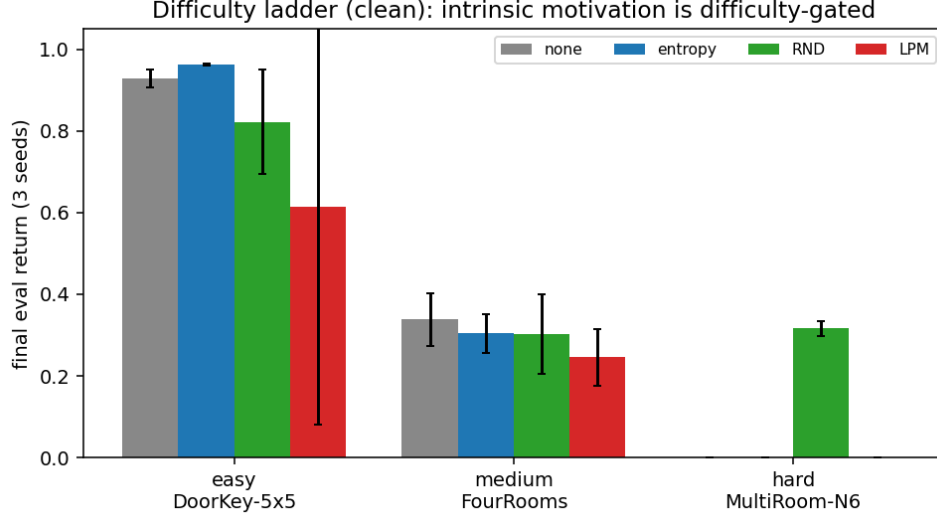


Figure 2: Difficulty ladder, clean, final eval return. Easy/medium: the baseline already solves/explores; intrinsic is unneeded (and slightly hurts). Hard MultiRoom-N6: only RND reaches the goal.

Why RND > LPM on clean hard exploration. Per-episode intrinsic magnitudes show the mechanism: RND’s bonus is *always positive* (mean = abs), a consistent directional drive toward novel states; LPM’s bonus is *signed* and nets ≈ 0 per episode (mean +0.08 vs. abs 0.33), giving no coherent exploration push — so the LPM agent behaves like the baseline and never explores out. Raising β scales LPM’s oscillation amplitude, not its directionality, so no β rescues it. This is LPM’s designed trade-off: its learning-progress signal goes to 0 both for unpredictable noise (its robustness virtue, next section) *and* for already-learned local dynamics (so it under-explores clean hard tasks).

4 Noise robustness: LPM vs. RND (RQ4)

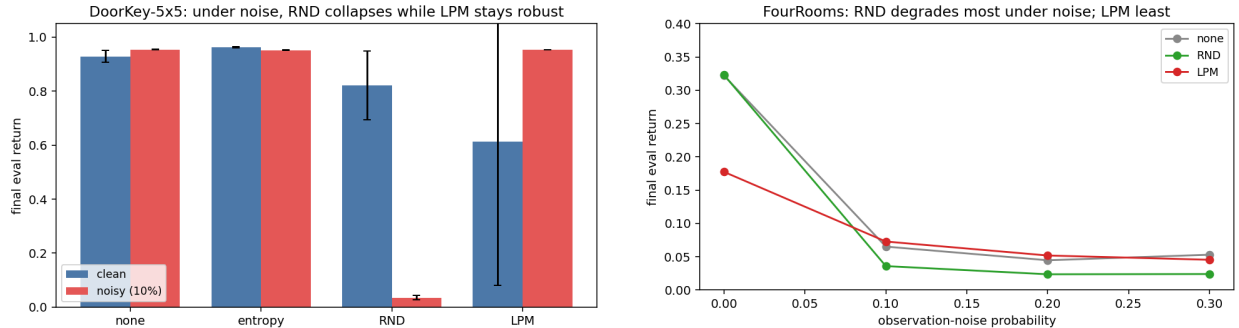


Figure 3: **Left:** DoorKey-5x5 clean vs. 10% observation noise — RND collapses (0.82 $\rightarrow$ 0.035), LPM stays robust (0.61 $\rightarrow$ 0.95, like the baseline). **Right:** FourRooms degradation vs. noise probability — RND falls off the most, LPM the least.

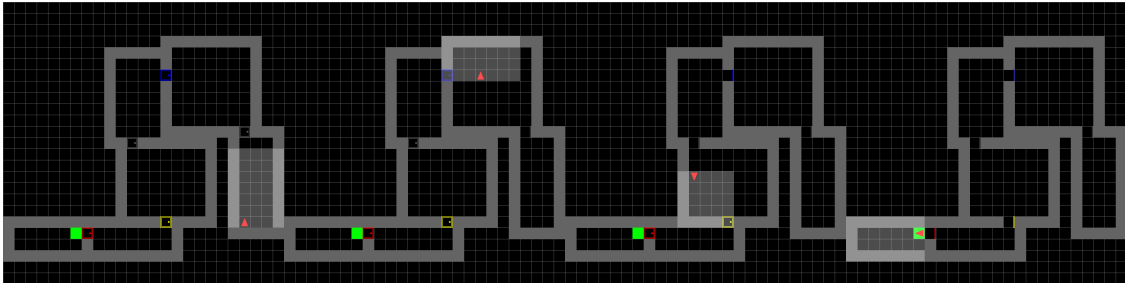
DoorKey-5x5	clean	noisy (10%)
NONE	0.93	0.95
ENTROPY	0.96	0.95
RND	0.82	0.035
LPM	0.61	0.95

Table 2: The cleanest noise-robustness result. DoorKey-5x5 is small enough that 10% noise does not stop NONE/ENTROPY/LPM from solving — which *isolates* the intrinsic-reward noise-vulnerability: only RND is destroyed.

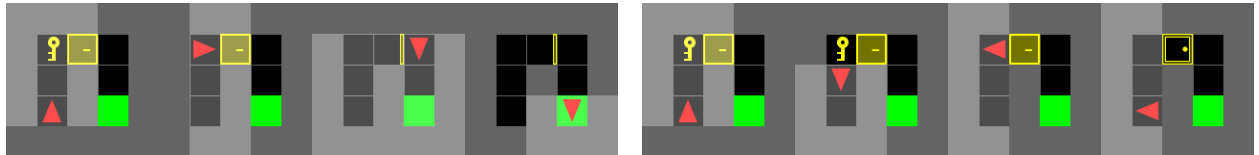
Observation noise makes every state look “novel,” so RND’s prediction-error bonus stays high and the agent chases noise instead of the goal; LPM’s learning-progress signal stays near 0 on unlearnable noise, so LPM is not distracted and effectively reverts to the (working) base policy. DoorKey-5x5 (Table 2) isolates this cleanly because the task is easy enough that noisy perception alone does not break it. On FourRooms the same *ordering* holds (RND hurt most, LPM least, Fig. 3 right) but the global noise also degrades the policy’s perception, so all methods fall to ≈ 0.05 — “LPM degrades less,” not “LPM stays solved.” On hard MultiRoom-N6, noise breaks perception for everyone (RND’s clean $0.32 \rightarrow 0$). *Slides 3.4/3.6.*

5 Agent traces (slide 3.5)

Animated GIFs live in `expr_data/minigrid/figures/traces/`; representative film-strips below.



Hard exploration — RND threads through the MultiRoom-N6 chain to the goal (reward 0.54).



Noise robustness on DoorKey-5x5 (10% noise): LPM (left) solves in ~ 13 steps; RND (right) wanders the full budget and fails.

Figure 4: Trajectory film-strips (evenly-spaced frames). Top: intrinsic motivation enables hard exploration. Bottom: LPM is noise-robust where RND is not.

6 Conclusions — answers to the research questions

- RQ1** *What do the LPM-paper experiments say about exploration metrics?* Coverage (MiniWorld) is a poor metric in extrinsic-free settings (a uniform-random policy maximises it). In MiniGrid we use *task success* / *return*, which actually discriminates exploration quality.
- RQ2** *Correct way to mix extrinsic/intrinsic; β too small/large?* β too large drowns the sparse reward ($\rightarrow 0$); too small is ignored. Usable ~ 0.001 – 0.005 , environment-dependent; on hard tasks there is a sweet spot (≈ 0.005 for RND) between under-exploring and never-converging (Fig. 1).
- RQ3** *Does intrinsic reward beat baselines on sparse MiniGrid?* Only when exploration is hard: **yes, decisively, on MultiRoom-N6 (RND 0.32 vs. 0.0)**; no on easy/medium where the baseline suffices (Fig. 2). RND > LPM on clean hard exploration.
- RQ4** *Does LPM generalise to other noise (beyond the MiniWorld noisy-TV)?* Yes — **LPM is noise-robust and RND noise-vulnerable**, cleanest on DoorKey-5x5 (RND 0.82 $\rightarrow$ 0.035, LPM $\rightarrow$ 0.95; Table 2). This reproduces the paper’s claim in a second, symbolic-observation domain.

One-line story for the deck: *intrinsic motivation buys you hard-exploration ability (RND) at the price of noise-fragility; LPM trades exploration aggressiveness for noise-robustness — and β must be tuned or it drowns the task.*

Slide-deck mapping (SPEC §3 “Intrinsic Reward and LPM”)

- 3.1 Environments / 3.2 Model & setup $\rightarrow$ §1.
- 3.3 β sweep $\rightarrow$ §2, Fig. 1.
- 3.4 Success rate across envs (clean and noisy) $\rightarrow$ §3–§4, Fig. 2, Table 1, Fig. 3.
- 3.5 Trace demonstrations $\rightarrow$ §5, Fig. 4.
- 3.6 LPM vs. RND, clean and noisy performance gap $\rightarrow$ §3 (clean: RND>LPM) + §4 (noisy: LPM>RND) — the rank flip is the punchline.

A Reproducibility & data

Code: `minigrid_exp/` (vendored + extended from Youssef’s `minigrid_intrinsic_reward`). Raw data, tables, figures and trace GIFs: `expr_data/minigrid/` with a running log in `FINDINGS.md`. **Compute note:** the 128-core box SIGKILLS any single process after ~ 18 min, so long runs use chunked checkpoint-resume training (`train_one --chunk-steps`, resume via `PP0.load`) driven by a meta-loop re-invoking `run_grid.py` (`ThreadPoolExecutor`); each chunk finishes under the limit and the idle driver survives. Grids: `run_grid.py --envs ... --methods rnd lpm --betas ... --noise-probs ... --seeds 1 2 3`. **Caveats:** (a) the MiniGrid noise is *global* per-element corruption, not a localised noisy-TV distractor; a localised variant would be a cleaner analogue. (b) LPM under-explores clean hard tasks here, so the clean LPM-vs-RND gap is RND-favourable; the noise result is the LPM-favourable half. (c) 3 seeds — DoorKey-clean LPM is bimodal (high variance).